Wazuh Home Lab

Wazuh is a security platform that combines SIEM (Security Information and Event Management) and XDR (Extended Detection and Response) capabilities.

In this lab, Wazuh will be used as a SIEM tool to monitor Windows 10 endpoint and detect suspicious activity.

Before starting the lab, it is important to understand the platform's four main components:

Component	Role
Wazuh Agent	Installed on monitored endpoints to collect and forward events and log to the server. In this lab, the agent is installed on the Windows 10 virtual machine.
Wazuh Server	Analyzes incoming events using decoders and detection rules to identify suspicious activity and generate security alerts.
Wazuh Indexer	Indexes and stores security alerts and other data, making them easier to search and retrieve.
Wazuh Dashboard	Provides graphics for visualizing alerts

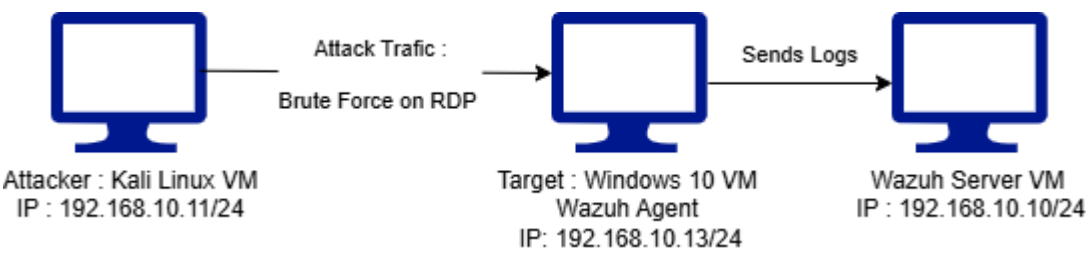
This lab explores two key Wazuh capabilities:

- File Integrity Monitoring (FIM): detect file and directory changes, including creation, modification, and deletion.

- Active Response: Configure an automated action when defined detection conditions are met.

The lab was completed over two separate days, so screenshots may show different dates and times

1.Environment Setup:

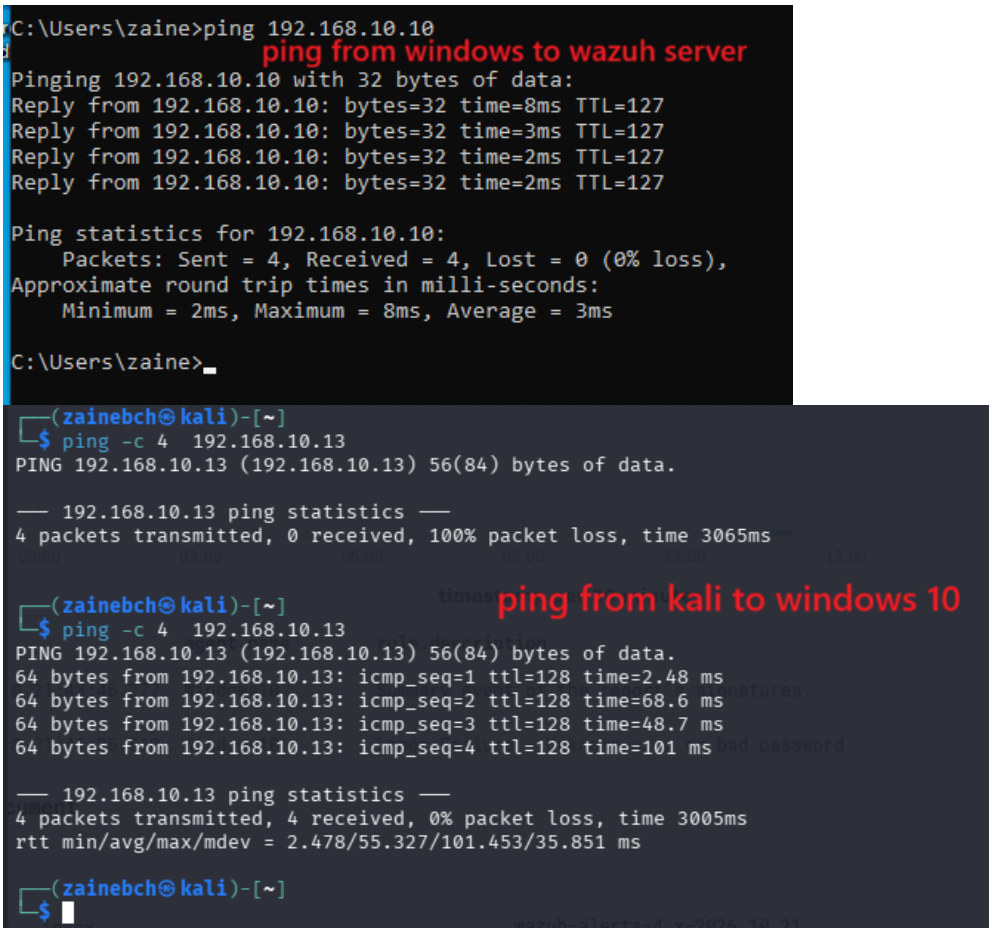


This lab was hosted in Oracle VM VirtualBox:

Component	Version	IP Address
Wazuh Server	Pre-built OVA — 4.14.9	192.168.10.10
Endpoint	Windows 10 Version 22H2	192.168.10.13
Kali Linux	2026.3	192.168.10.11

- Configure static IP addresses within the 192.168.10.0/24 network

-Confirm connectivity



At first, ping requests from Kali Linux to Windows 10 VM failed because Windows firewall was blocking the traffic. So, I disabled It.

It is acceptable to do this since it is a lab environment, but not in a production environment.

-Add windows endpoint: install the Wazuh Agent on the Windows 10 VM and activate Wazuh server

```
PS C:\Windows\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.9-1.msi -OutFile
$env:tmp\wazuh-agent; msiexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='192.168.10.10' WAZUH_AGENT_NAME='Windows10'
PS C:\Windows\system32> NET START WazuhSvC
The Wazuh service is starting.
The Wazuh service was started successfully.

PS C:\Windows\system32>
```

-Access to Wazuh GUI through Kali's web browser https://192.168.10.10 (Wazuh Server IP address)

2. Windows Account Activity and Event Monitoring:

a. Creating a new user

```
S C:\Windows\system32> net user TestUser Password123 /add
he command completed successfully.

S C:\Windows\system32> net user

ser accounts for \\DESKTOP-E2G35HE

-------------------------------------------------------------------------------
dministrator            DefaultAccount           defaultuser100000
efaultuser100001        Guest                    TestUser
DAGUtilityAccount       zaine
he command completed successfully.

S C:\Windows\system32>
```

On Wazuh a security event with ID 4720 ("A user account was created") was recorded confirming the creation of the user TestUser.

t data.win.system.eventID	4720
t data.win.system.eventRecordID	7908
t data.win.system.keywords	0x8020000000000000
t data.win.system.level	0
t data.win.system.message	"A user account was created. Subject: Security ID: S-1-5-21-1768321648-3769934198-3604349264-1001 Account Name: zaine Account Domain: DESKTOP-E2G35HE Logon ID: 0x6175F New Account: Security ID: S-1-5-21-1768321648-3769934198-3604349264-1005 Account Name: TestUser Account Domain: DESKTOP-E2G35HE Attributes: SAM Account Name: TestUser Display Name: <value not set> User Principal Name: - Home Directory: <value not set> Home Drive: <value not set> Script Path: <value not set> Profile Path: <value not set> User Workstations: <value not set> Password Last Set: <never> Account Expires: <never> Primary Group ID: 513 Allowed To Delegate To: - Old UAC Value: 0x0 New UAC Value: 0x15 User Account Control: Account Disabled 'Password Not Required' - Enabled 'Normal Account' - Enabled User Parameters: <value not set> SID History: - Logon Hours: All Additional Information: Privileges -"

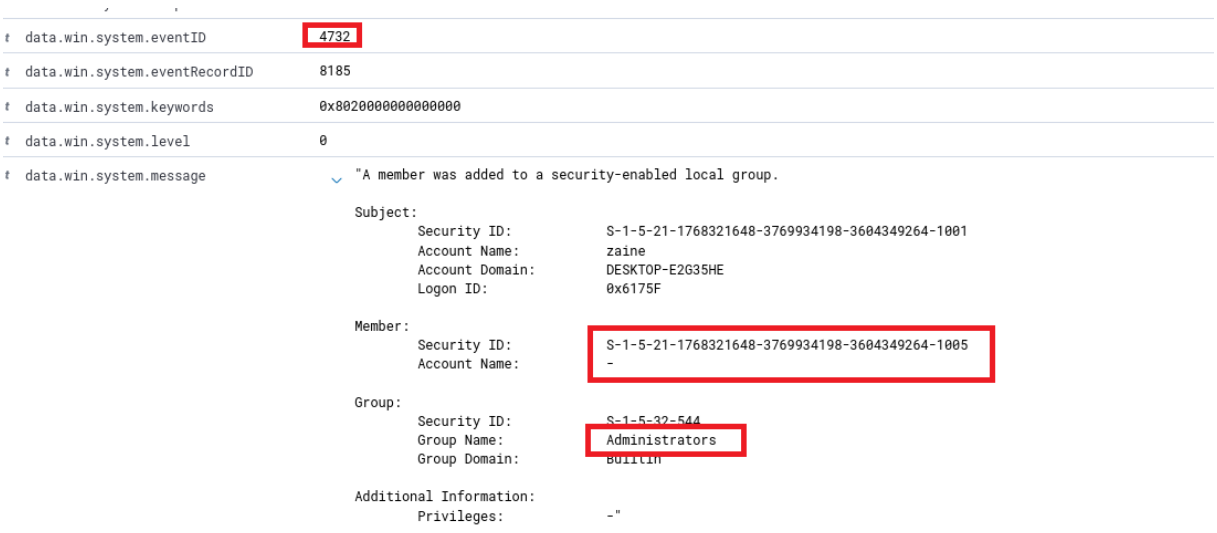
It is important to note down the security ID, as it identifies the user object.

Security ID: S-1-5-21-1768321648-3769934198-3604349264-1005

b. Adding a user to a group

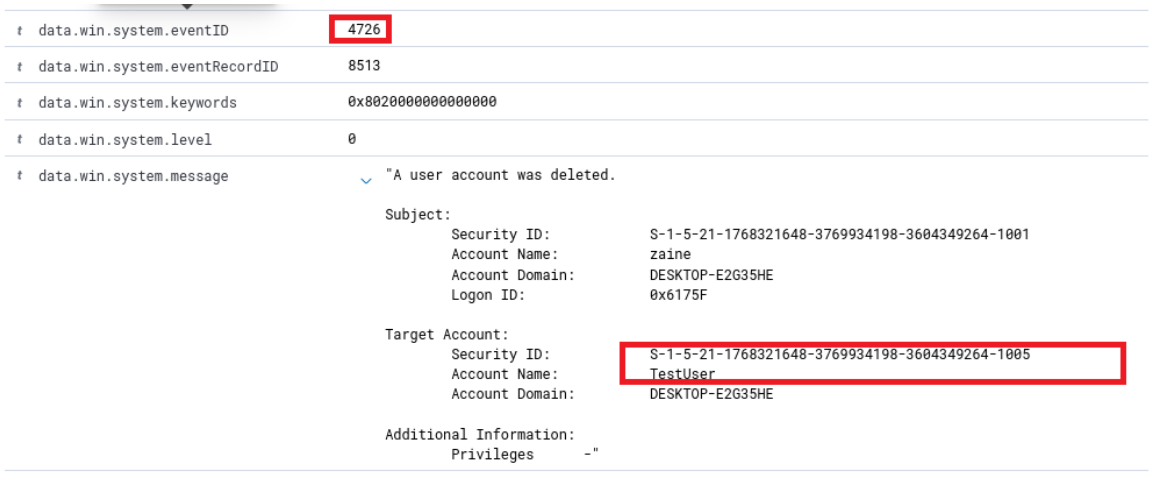
```
PS C:\Windows\system32> net localgroup Administrators TestUser /add
The command completed successfully.
```

A security event with ID 4732 ("A member was added to a security-enabled local group") was recorded on Wazuh.
In the event, account name is not specified, but security ID can used to identify that it was TestUser who was the one added to the Administrators group.



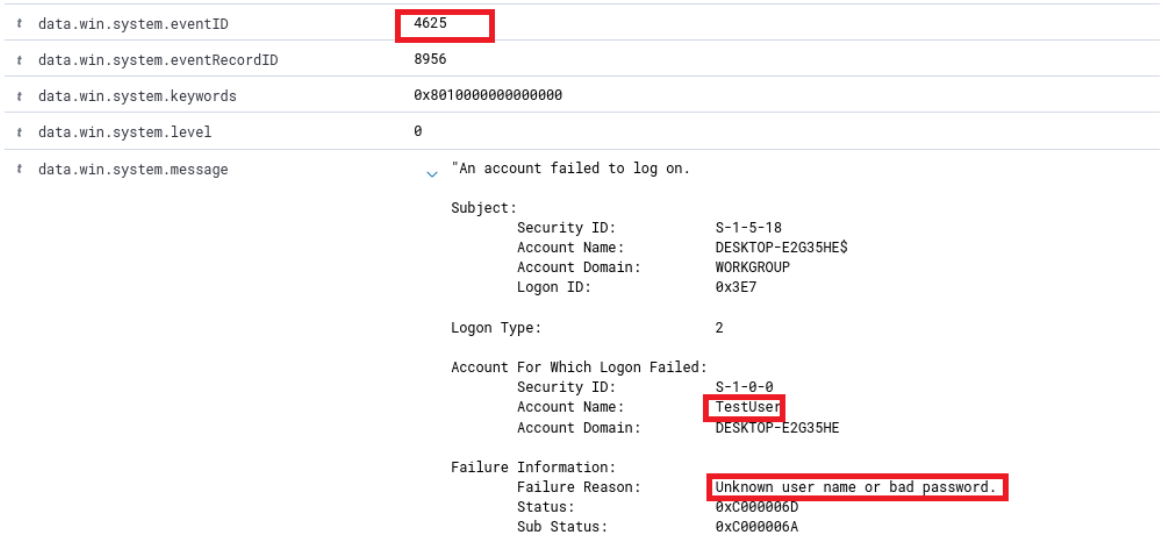
c. Deleting a user

```
PS C:\Windows\system32> net user TestUser /delete
The command completed successfully.
```

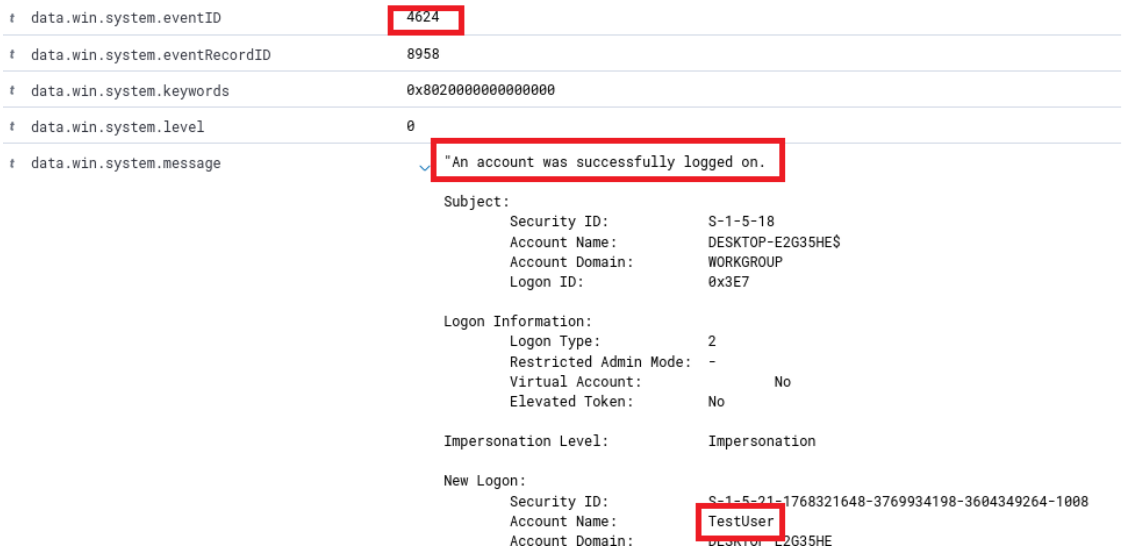


d. Logons

After typing a wrong password, an event with Event ID 4625 (“An account failed to log on.”) was recorded on Wazuh.



After typing the correct credentials, an event with Event ID 4624 (“An account was successfully logged on.”) was recorded on Wazuh**.**



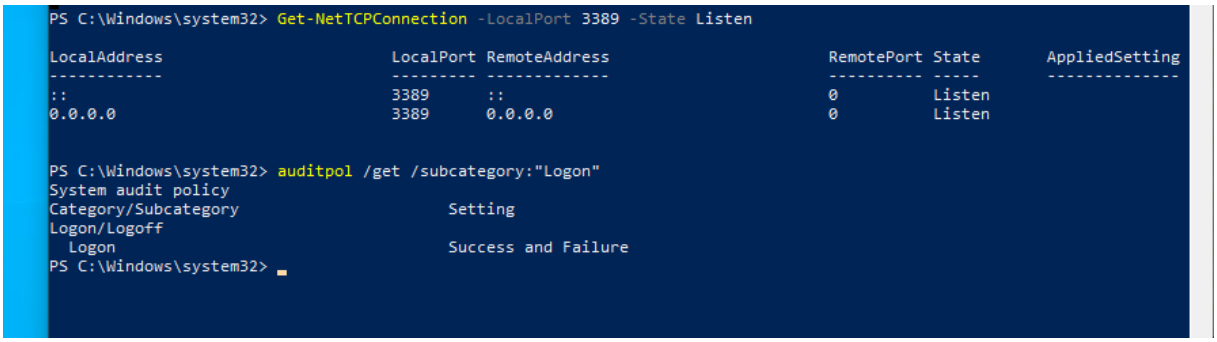
3.RDP Simulation Attack

RDP(Remote Desktop Protocol) is a protocol that allows you to connect to and control a Windows computer remotely.

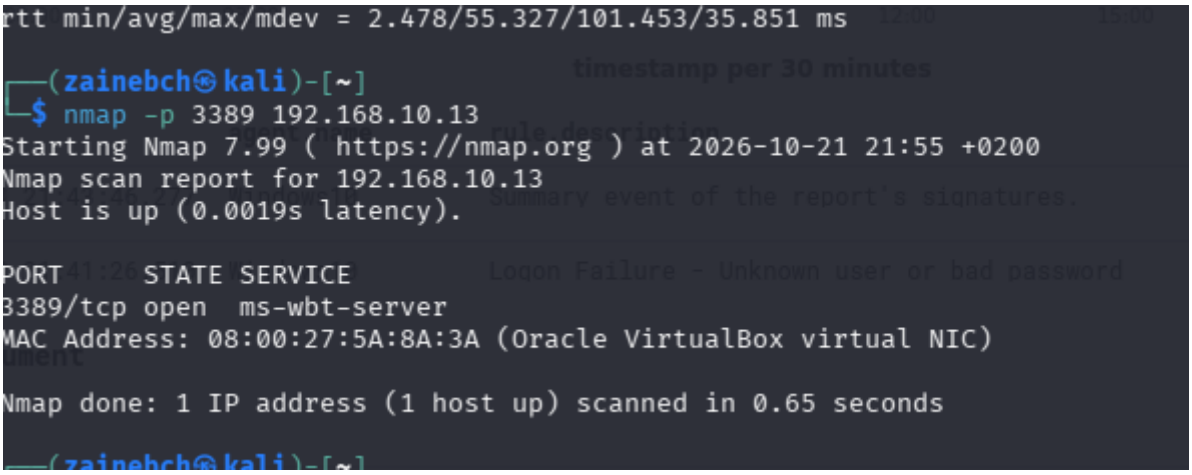
In this lab, RDP is the attack vector so first thing is to enable it on Windows 10 machine.

- Enable RDP on Windows VM, by default it is on port 3389

Verifying that port 3389 is in a listening state



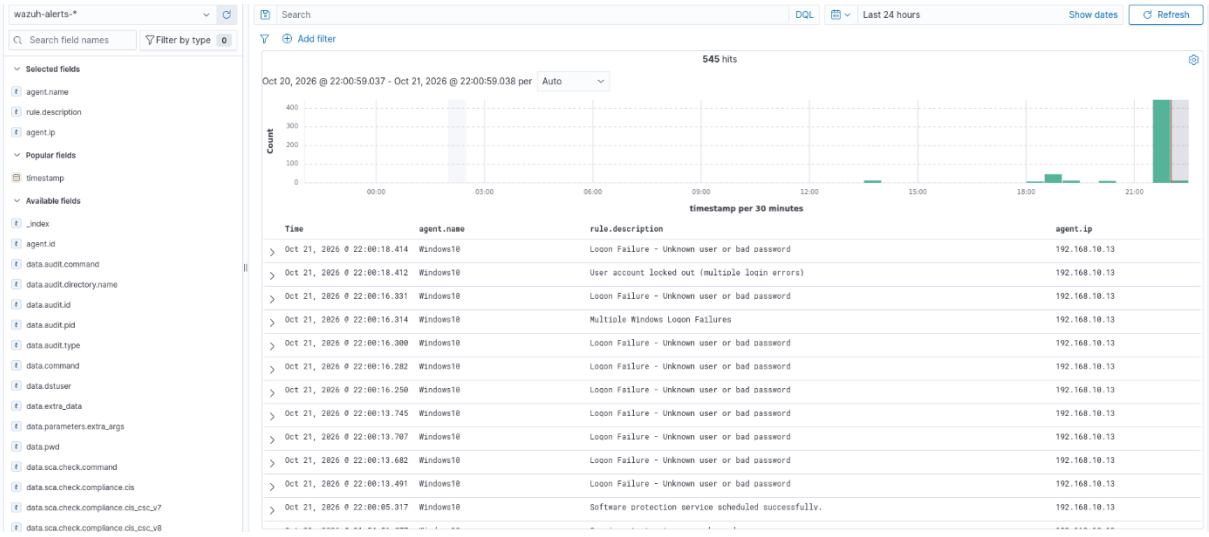
Nmap scan from Kali Linux confirming that the RDP service is open and accessible on windows VM



-On Kali Linux VM , launch an RDP brute force attack using Hydra against Windows 10 target's Remote Desktop service on user TestUser.

hydra -V -l TestUser -P passwords.txt rdp://192.168.10.13

Wazuh successfully collected endpoint telemetry showing repeated authentication failures and confirming the simulated attack launched from kali



- Event ID : 4625
- Target User: TestUser
- Source Workstation: kali
- Source IP: 192.168.10.11
- Triggered Alert: Rule 60204 (Multiple Windows Logon Failures)

Wazuh automatically mapped the correlated event telemetry to MITRE ATT&CK

- Tactic: Credential Access
- Technique: Brute Force (T1110)

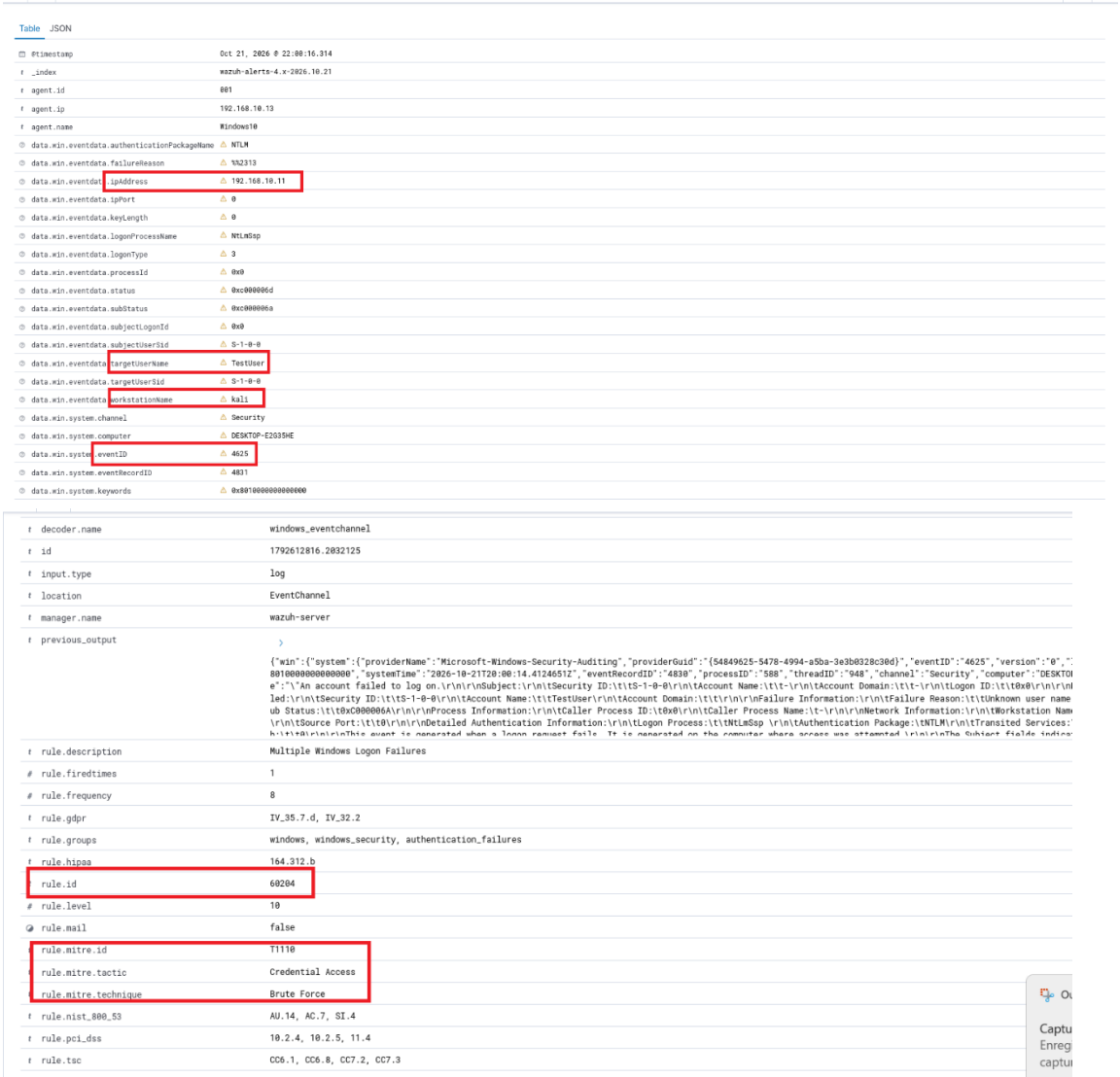


Table JSON	
@timestamp	Oct 21, 2026 @ 22:08:16.314
_index	wazuh-alerts-4.x-2026.10.21
agent.id	001
agent.ip	192.168.10.13
agent.name	Windows10
data.win.eventdata.authenticationPackageName	NTLM
data.win.eventdata.failureReason	%%2313
data.win.eventdata.ipAddress	192.168.10.11
data.win.eventdata.ipPort	0
data.win.eventdata.keyLength	0
data.win.eventdata.logonProcessName	NtLmSsp
data.win.eventdata.logonType	3
data.win.eventdata.processId	0x0
data.win.eventdata.status	0xc000006d
data.win.eventdata.subStatus	0xc000006a
data.win.eventdata.subjectLogonId	0x0
data.win.eventdata.subjectUserSid	S-1-0-0
data.win.eventdata.targetUserName	TestUser
data.win.eventdata.targetUserSid	S-1-0-0
data.win.eventdata.workstationName	kali
data.win.system.channel	Security
data.win.system.computer	DESKTOP-E2G35HE
data.win.system.eventID	4625
data.win.system.eventRecordID	4831
data.win.system.keywords	0x8010000000000000

decoder.name	windows_eventchannel
id	1792612816.2032125
input.type	log
location	EventChannel
manager.name	wazuh-server
previous_output	{"win":{"system":{"providerName":"Microsoft-Windows-Security-Auditing","providerGuid":"{54849625-5478-4994-a5ba-3e3b0328c30d}","eventID":"4625","version":"0","8010000000000000","systemTime":"2026-10-21T20:00:14.4124651Z","eventRecordID":"4830","processID":"588","threadID":"948","channel":"Security","computer":"DESKTO e":"\"An account failed to log on.\r\n\r\nSubject:\r\n\tSecurity ID:\t\tS-1-0-0\r\n\tAccount Name:\t\t-\r\n\tAccount Domain:\t\t-\r\n\tLogon ID:\t\t0x0\r\n\r\n led:\r\n\tSecurity ID:\t\tS-1-0-0\r\n\tAccount Name:\t\tTestUser\r\n\tAccount Domain:\t\t\r\n\r\nFailure Information:\r\n\tFailure Reason:\t\tUnknown user name ub Status:\t\t0xC000006A\r\n\r\nProcess Information:\r\n\tCaller Process ID:\t0x0\r\n\tCaller Process Name:\t-\r\n\r\nNetwork Information:\r\n\tWorkstation Nam \r\n\tSource Port:\t\t0\r\n\r\nDetailed Authentication Information:\r\n\tLogon Process:\t\tNtLmSsp \r\n\tAuthentication Package:\tNTLM\r\n\tTransited Services:
rule.description	Multiple Windows Logon Failures
rule.firedtimes	1
rule.frequency	8
rule.gdpr	IV_35.7.d, IV_32.2
rule.groups	windows, windows_security, authentication_failures
rule.hipaa	164.312.b
rule.id	60204
rule.level	10
rule.mail	false
rule.mitre.id	T1110
rule.mitre.tactic	Credential Access
rule.mitre.technique	Brute Force
rule.nist_800_53	AU.14, AC.7, SI.4
rule.pci_dss	10.2.4, 10.2.5, 11.4
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3

Recommended Mitigations according to MITRE ATT&CK framework:

- Account Lockout Policy: Configure appropriate account lockout thresholds and durations to limit repeated password-guessing attempts.
- Multi-Factor Authentication (MFA)

4.File Integrity Monitoring

-Enable the FIM capability:

Open ossec.conf file on notepad with administrator privileges and in the file montoring n and add the directory you want to monitor.

```
<!-- File integrity monitoring -->
<syscheck>

  <disabled>no</disabled>
  <directories realtime="yes">C:\Secret</directories>
```

Restart wazuh service to apply changes.

t decoder.name	syscheck_new_entry
t full_log	File 'c:\secret\data.txt' added Mode: realtime
t id	1792925373.817687

t decoder.name	syscheck_deleted
t full_log	File 'c:\secret\data.txt' deleted Mode: realtime
t id	1792925348.814607

t full_log	File 'c:\secret\data.txt' modified Mode: realtime Changed attributes: size,mtime,md5,sha1,sha256 Size changed from '7' to '13' Old modification time was: '1792921771', now it is '1792921968' Old md5sum was: '2ee62715b696c1801533d52caad8ba64' New md5sum is : '13beb619235e412d5c52fe3c99e7270c' Old sha1sum was: 'd1ebbc78705ad98aa2c8fb357d2e30c8f001673a' New sha1sum is : 'd00cb2d33a4b74026576f6f60af6929b81187181' Old sha256sum was: 'c34f63ae5db1f9d5eff9f174617fd21cfdd8450d22432eac0db334c80c355370' New sha256sum is : '9dfc5fa0c0e9b90b72953f383129c6a7c814ec5efd6ea8a45b162a5c5375943b'

4/ Active Response:

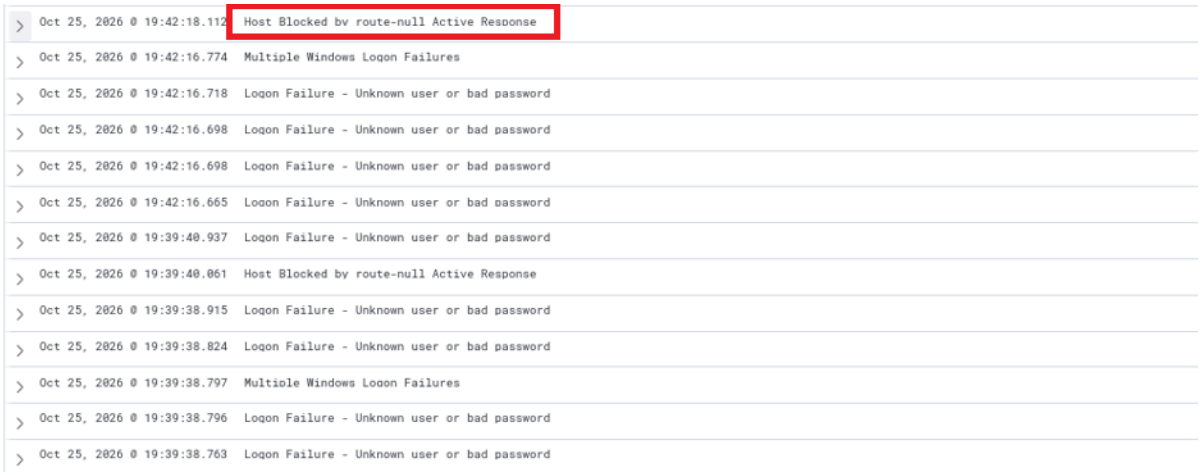
The idea is to configure Wazuh Active Response to automatically block traffic originating from the IP address source of the RDP brute-force attack.

-Use rule ID associated with the detected attack to trigger the response which is 60204 to configure on Wazuh Server :check Wazuh documentation

On /var/ossec/etc/ossec.conf on Wazuh server :

```
<active-response>
  <disabled>no</disabled>
  <command>win_route-null</command>
  <location>local</location>
  <rules_id>60204</rules_id>
</active-response>
```

When this rule is triggered, Wazuh executes a command to add a null route for the attacker's source IP address.
A null route discards packets destined for the specified IP address, preventing further communication



References :

MyDFIR Wazuh SOC Analyst Challenge - YouTube

Wazuh Documentation